

THE CLONE WARS



A deep dive into biometric attacks, adversarial techniques and main defences

SECFORCE

Agenda

1. Intro
2. Presentation/Injection Attacks - Face and Voice
3. Adversarial Techniques and Laundering Attacks
4. Remediations
5. Q&A

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Introduction

$whoami



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Biometric Security



AI/LLM Security



Penetration Testing



In the News



Reuters

My News

AI messaging scam costs Italy's top bank Intesa millions, sources say

By Emilio Parodi

BBC

Cloned customer voice beats bank security checks

28 November 2024

Shari Vahl

Reporter, You and Yours

CNN World

WORLD > ASIA • 2 MIN READ

Finance worker pays out $25 million after video call with deepfake 'chief financial officer'

FEB 4, 2024

By Heather Chen, Kathleen Magramo

Industry News

Resultsense

8 June 2026

2 min read

via The Guardian

Aviva detects record £233m in fraud as AI scams rise

Aviva flagged a record £233m in suspect claims in 2025 as fraudsters used AI to fake accidents and documents, prompting the insurer to deploy AI in defence.

Attack Scenarios

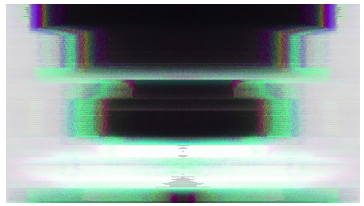
Enrolment

Identity Concealment

Goal: hide or fabricate the true identity so it cannot be linked or de-duplicated.



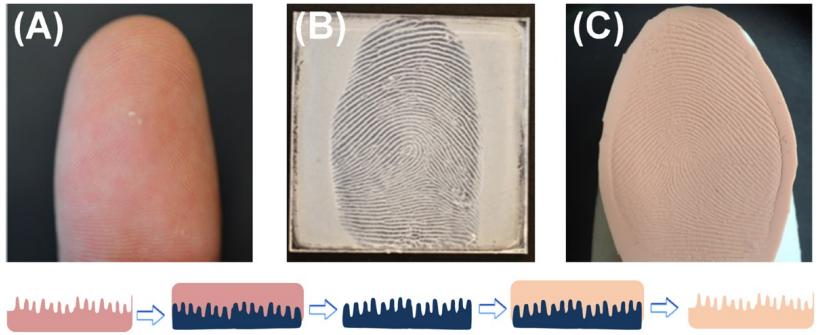


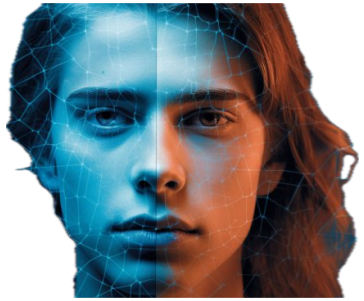


Authentication

Identity Impersonation

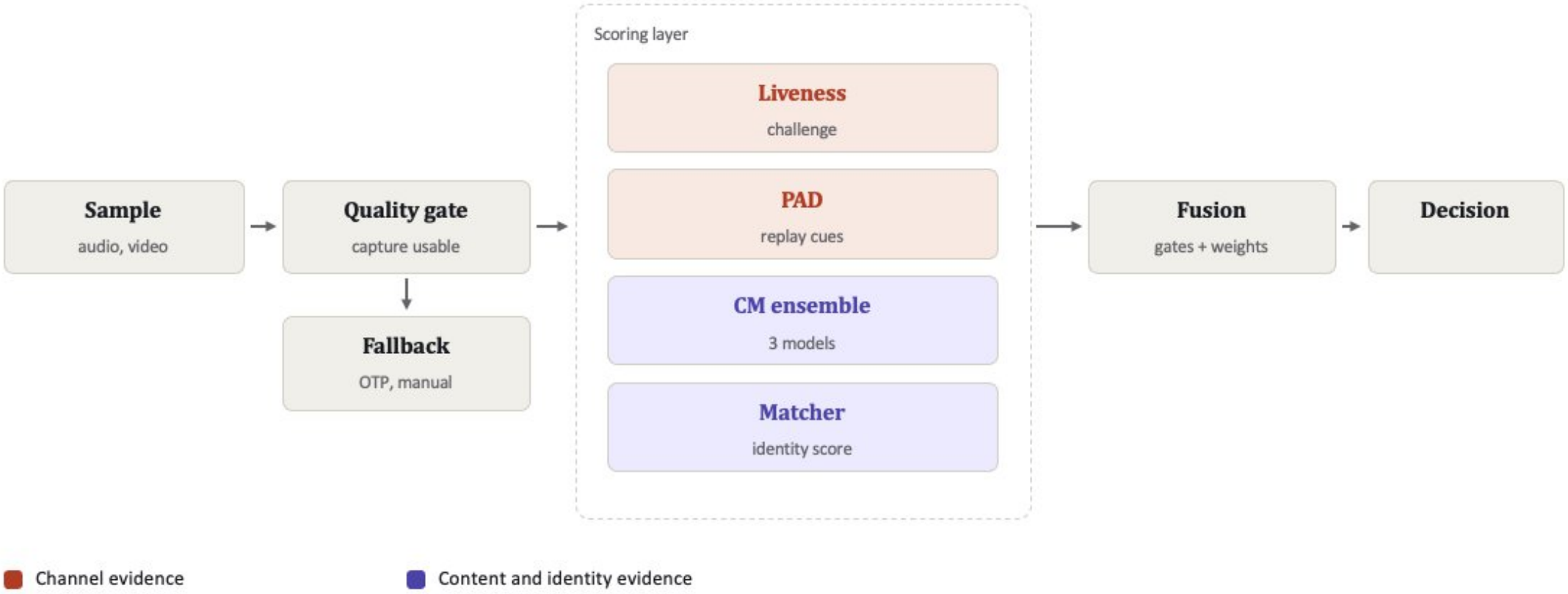
Goal: be verified as one specific enrolled victim → account takeover.



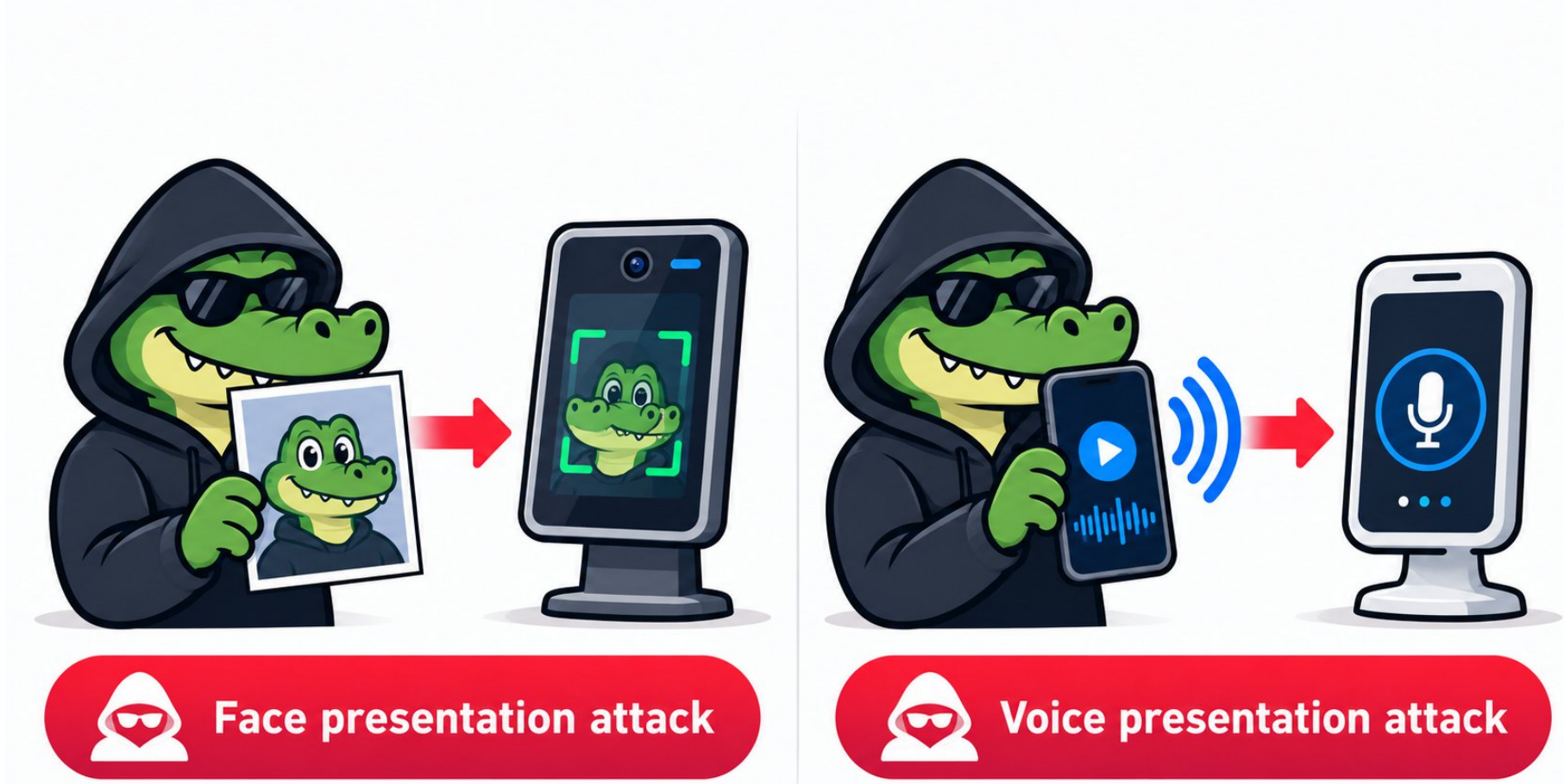




Biometric Authentication Process



Presentation Attacks - Introduction



Presentation Attack Instruments







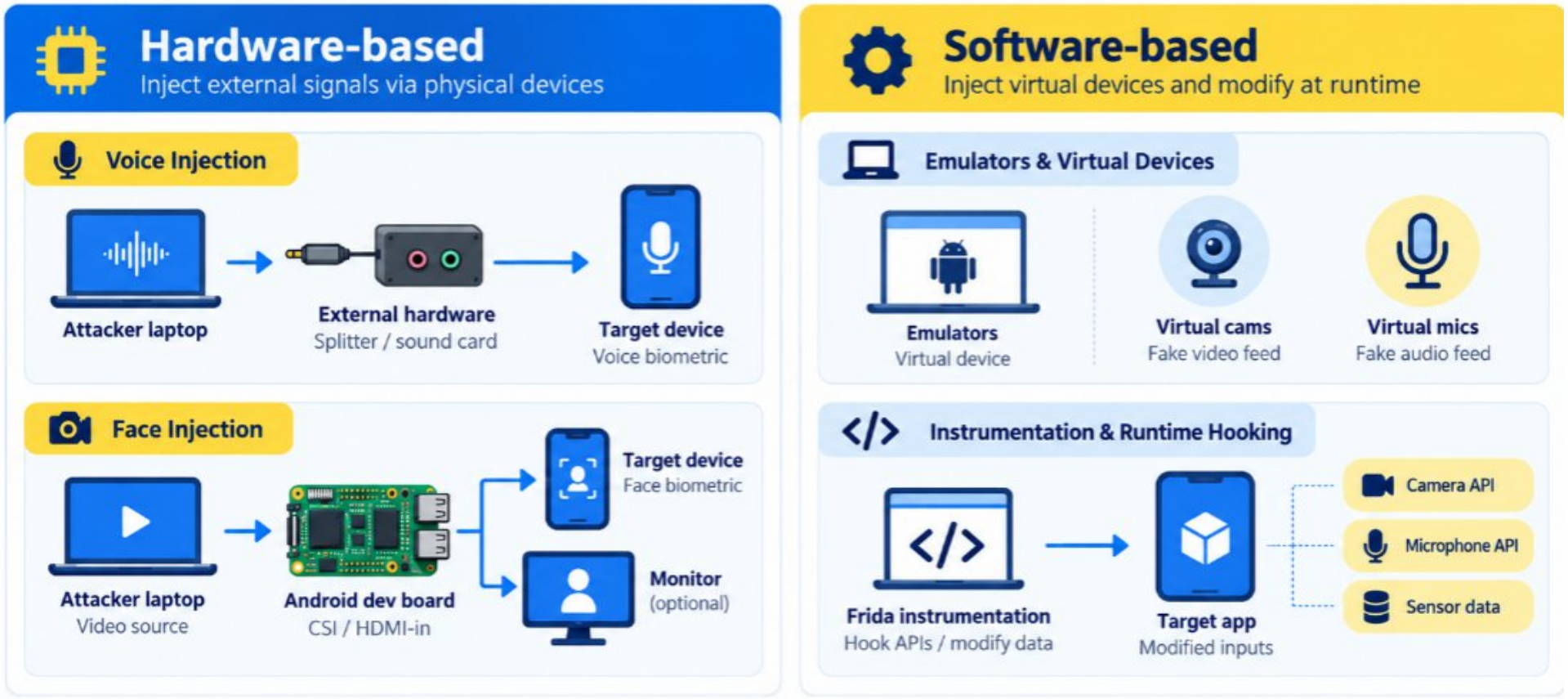








Injection Attack Types



Attack Toolchain – Hardware Audio Injection

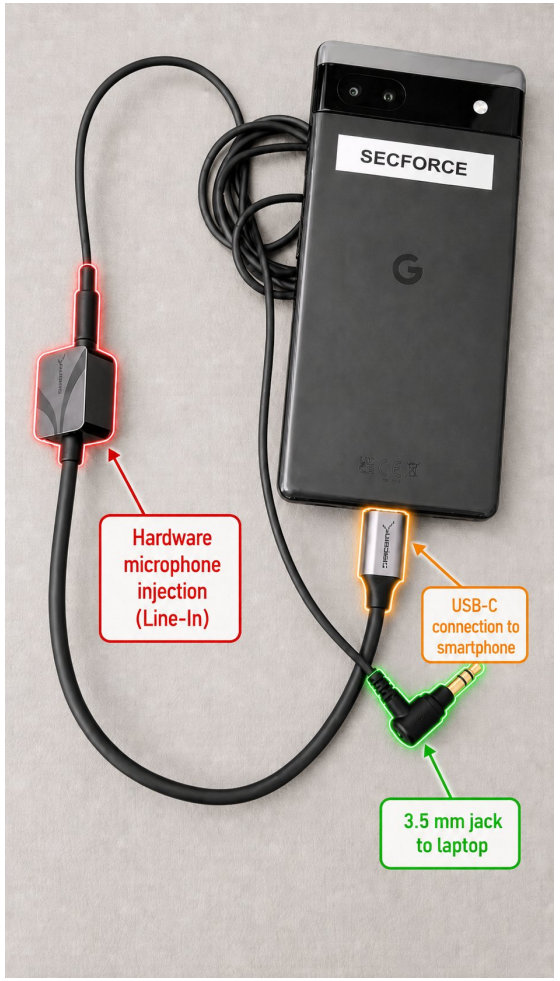
A few years ago...



(a) For Android Devices (b) For iOS Devices

Biometric Authentication Under Threat: Liveness Detection Hacking, BlackHat USA 2019

Today!



IVR Systems and Audio Injection



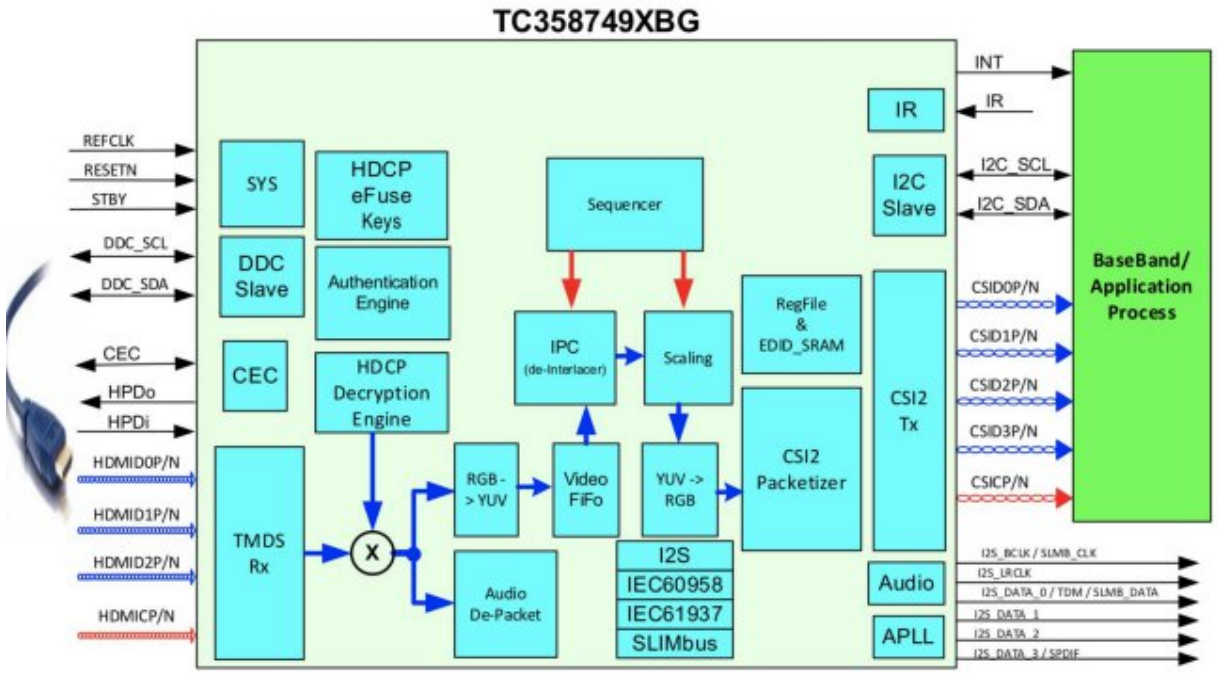
IVR system Audio Injection Process

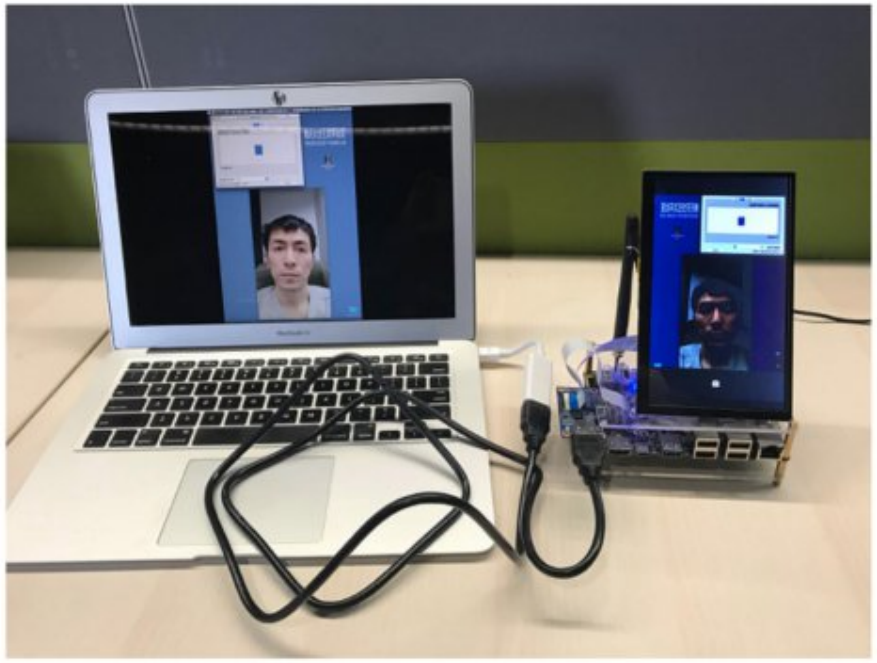


Attack Toolchain – Hardware Video Injection

A few years ago… → **Today?**

Video Injection Example

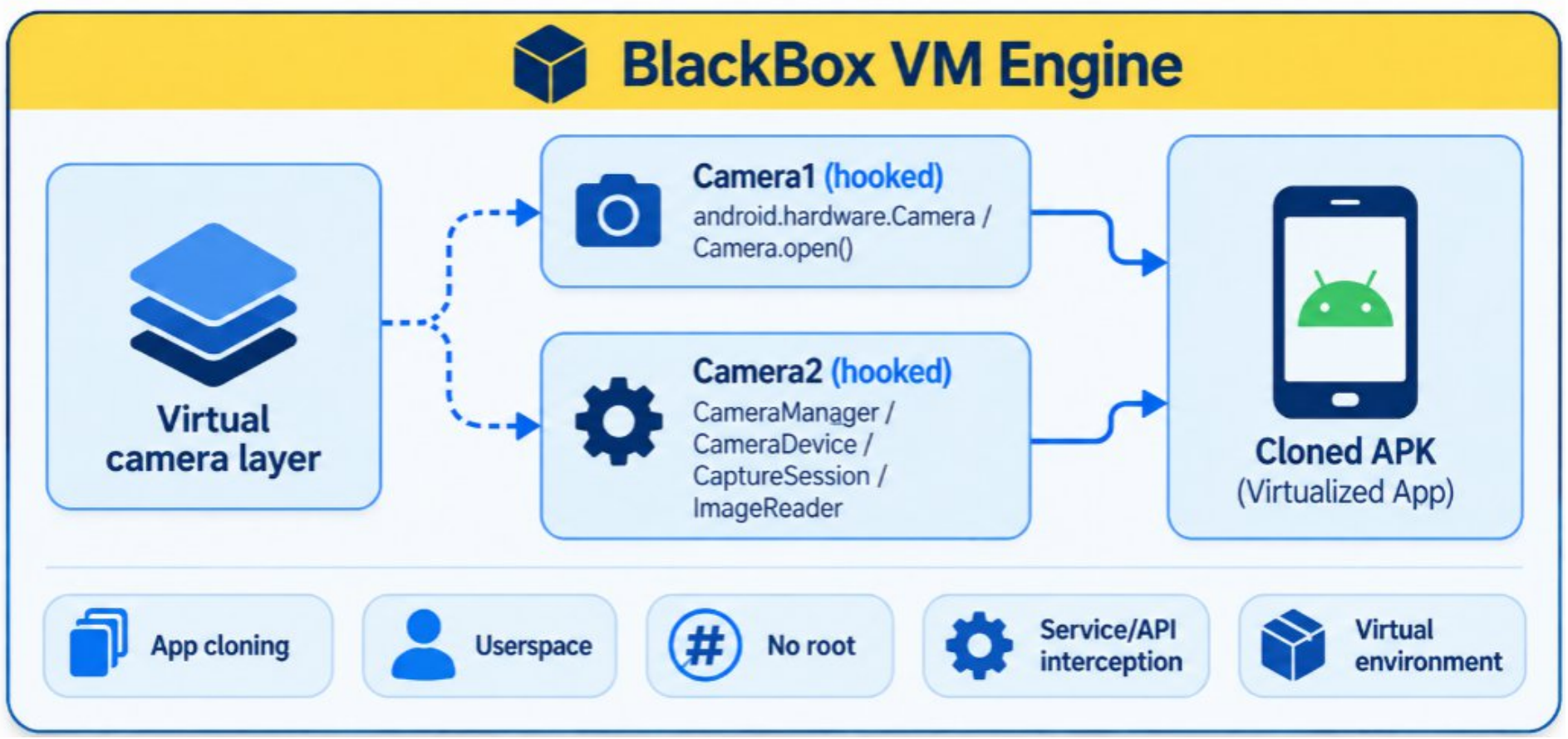




Video injection device based on TC358749XBG

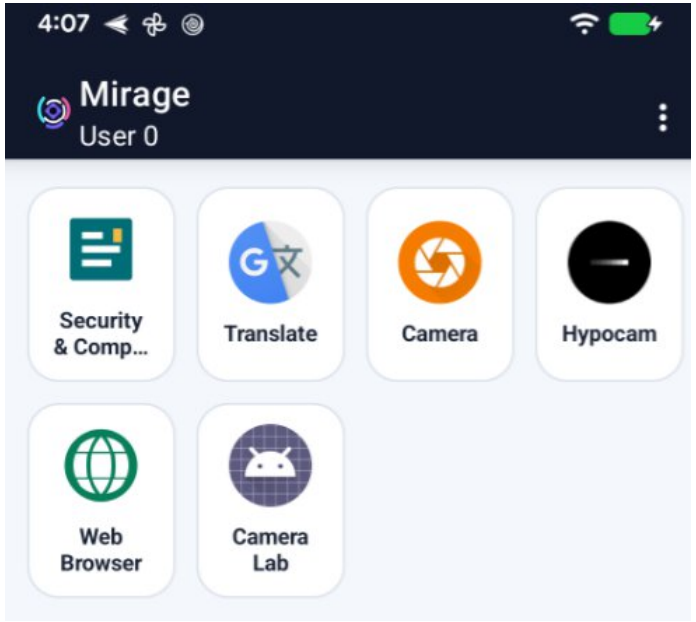
Biometric Authentication Under Threat: Liveness Detection Hacking, BlackHat USA 2019

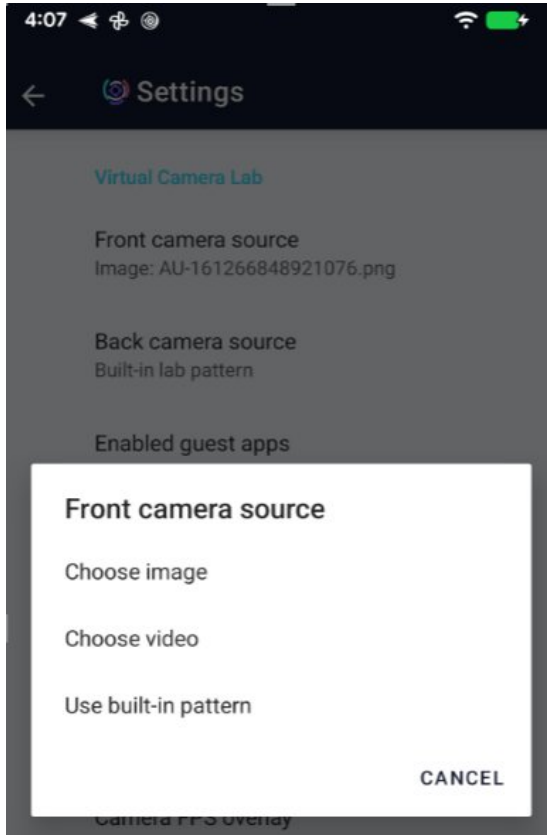
Android Camera Injection - Mirage App

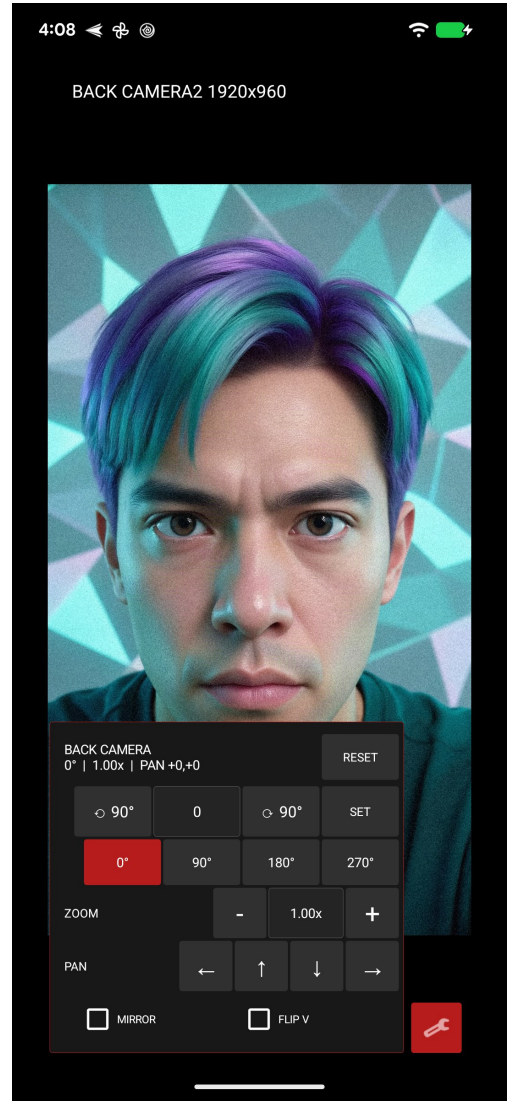


SECFORCE custom injection app based on the BlackBox engine —-> https://github.com/FBlackBox/BlackBox

Android Camera Injection - Mirage App

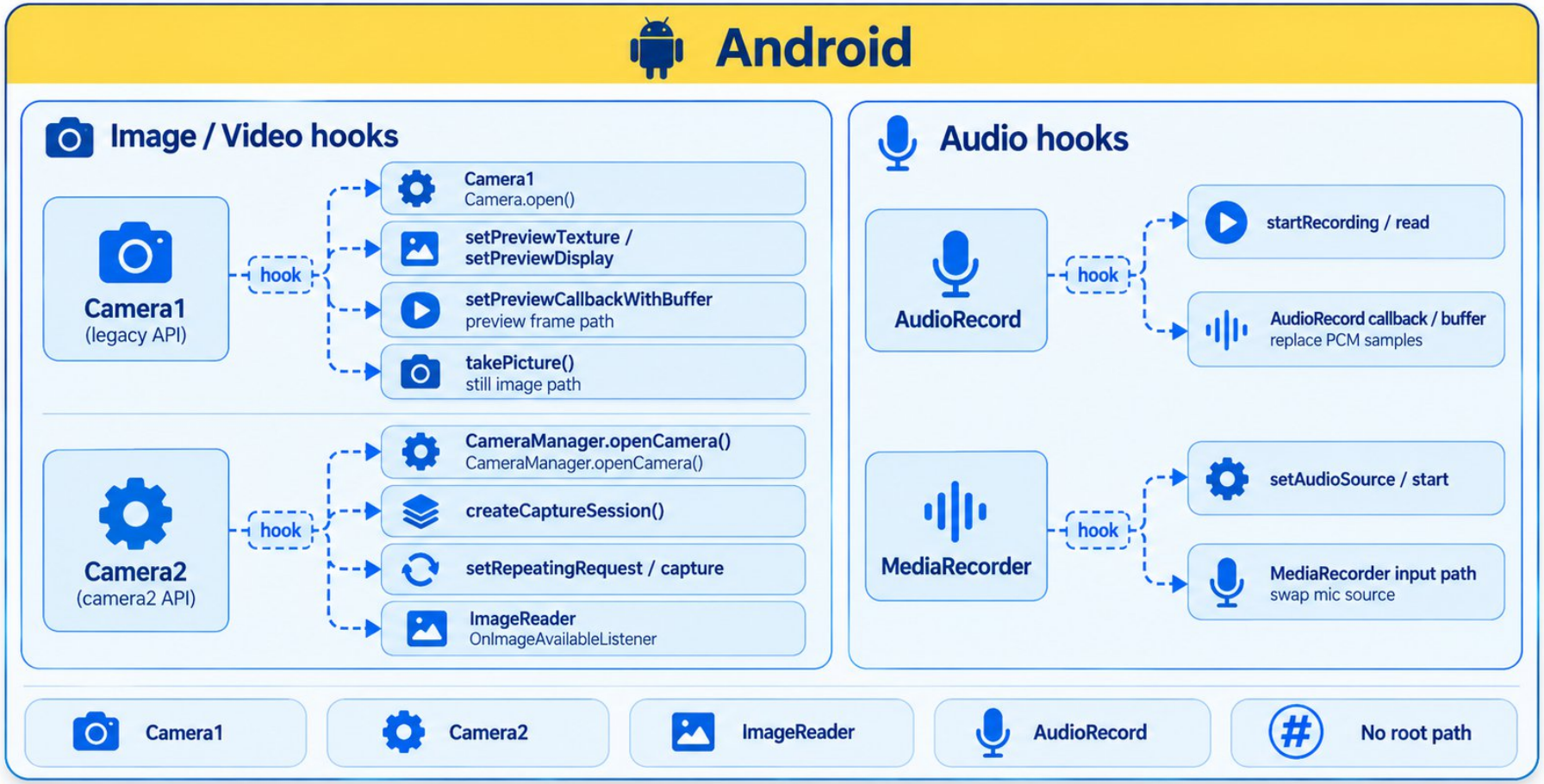




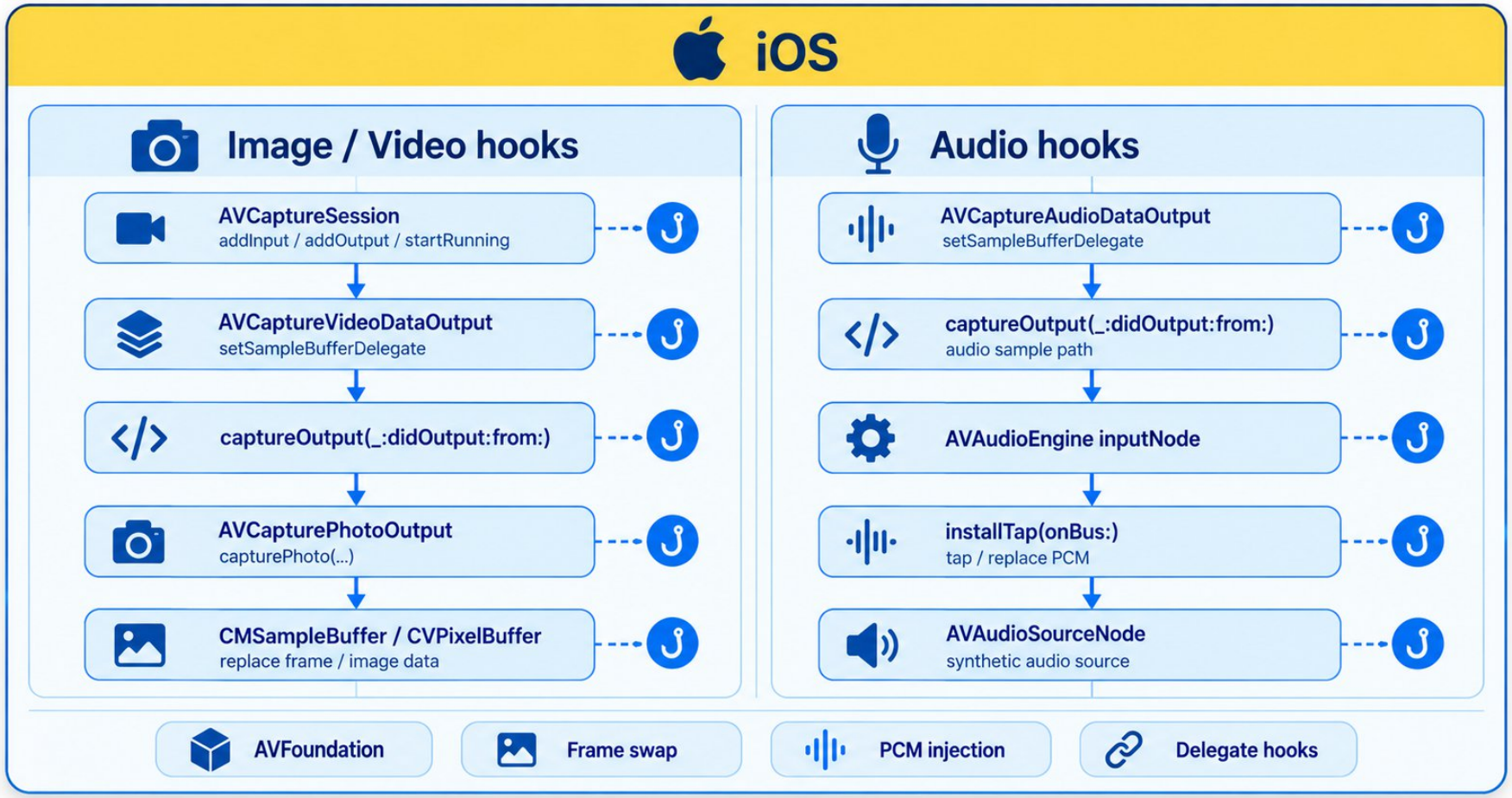


DEMO

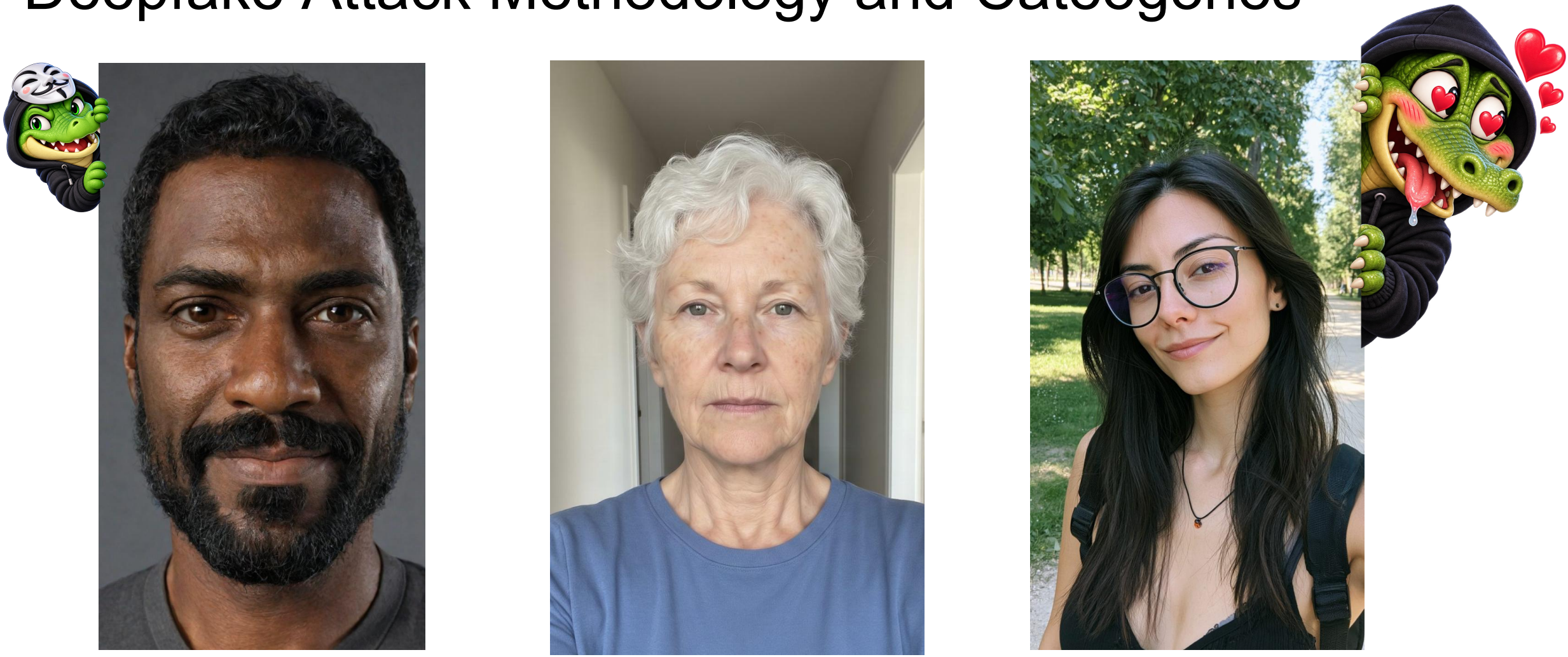
Android Biometric Injection - Frida



iOS Biometric Injection - Frida

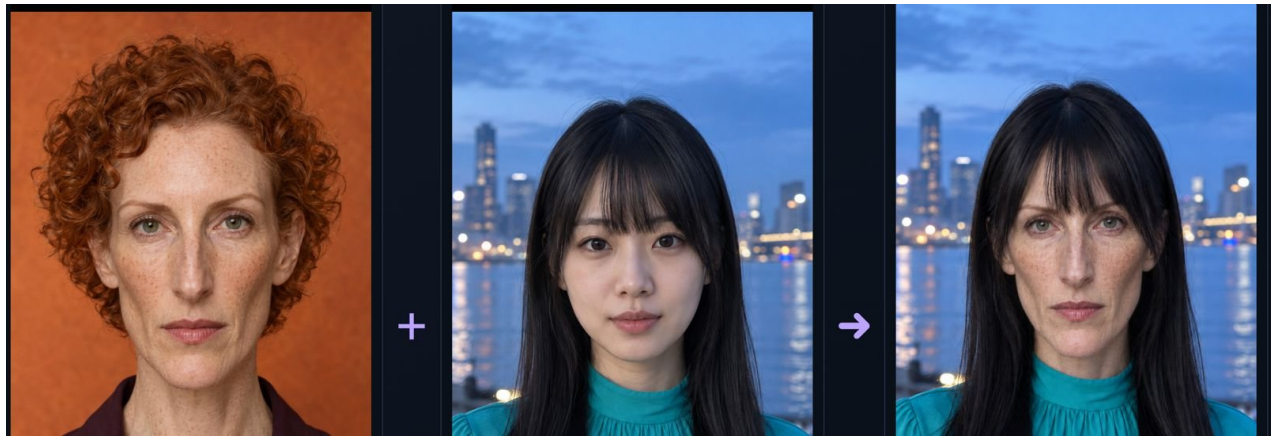


Deepfake Attack Methodology and Cateogories



Deepfake Attack Methodology and categories

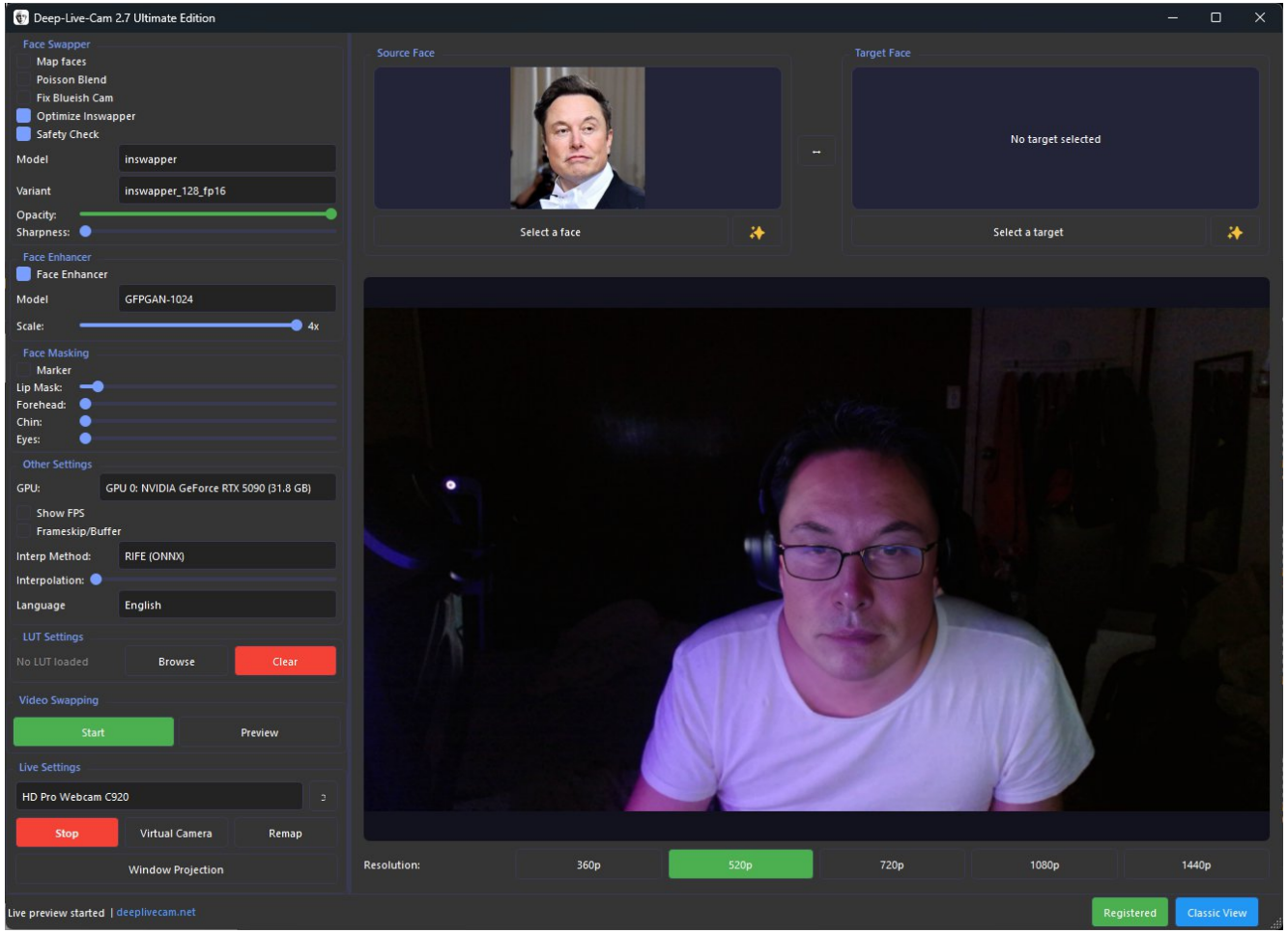
Face Swap





Face Morph

Deepfake Attack Methodology and cateogories



DeepLive Cam

https://github.com/hacksider/deep-live-cam

Deepfake Attack Methodology and cateogories

Sunglasses + rotated head



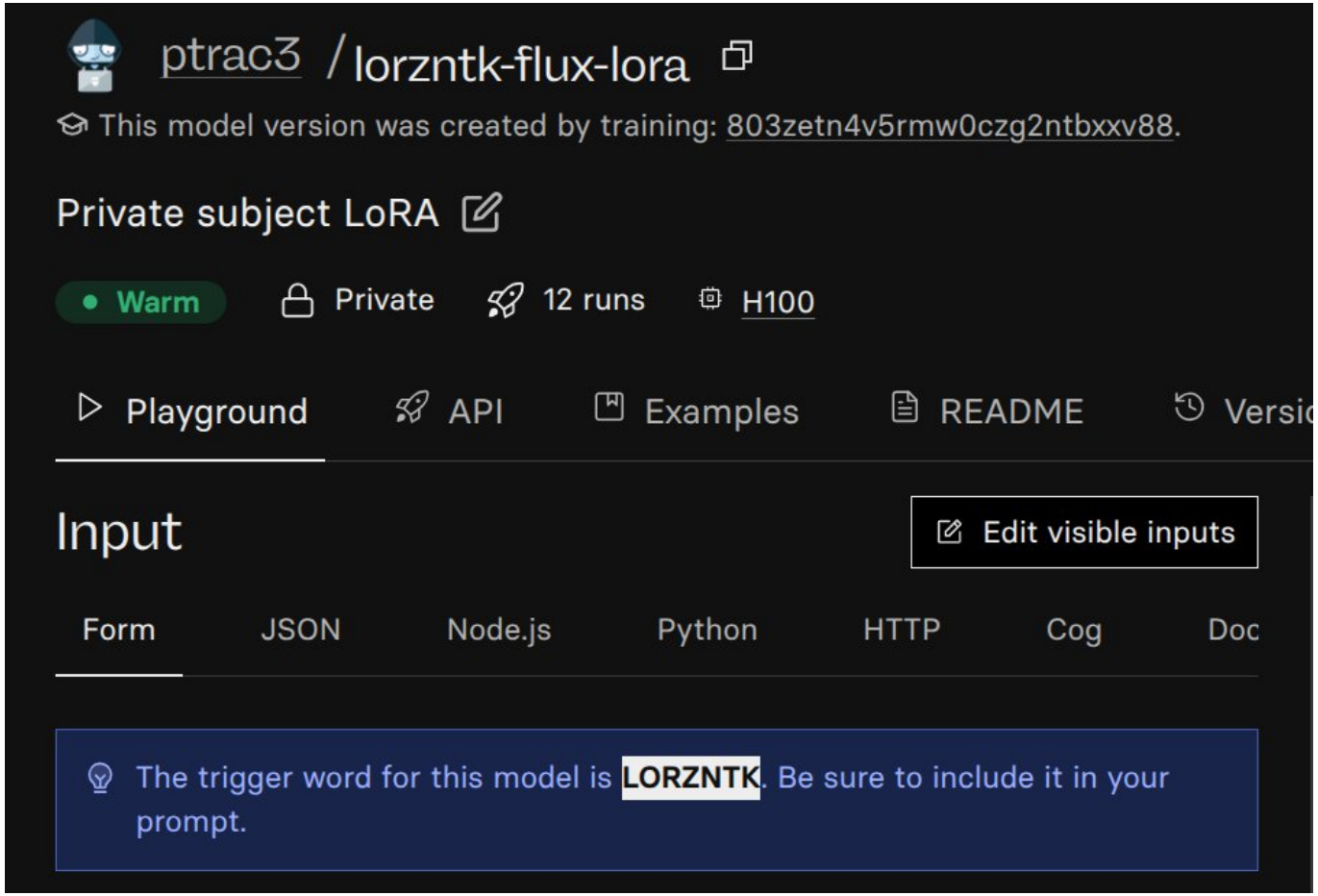




AI-generated frontal reconstruction

Image-editing Gemini Pro model

Deepfake Attack Methodology and cateogories



Identity Fine-tuning via Replicate AI Flux Dev 2 LoRA training

- **Generation of 20/30 samples, same subject, different lighting/background/clothing**
- **A unique trigger word allows to force the model to generate images using the previously ingested dataset as the reference**
- **Prompt engineering for the best-looking images**

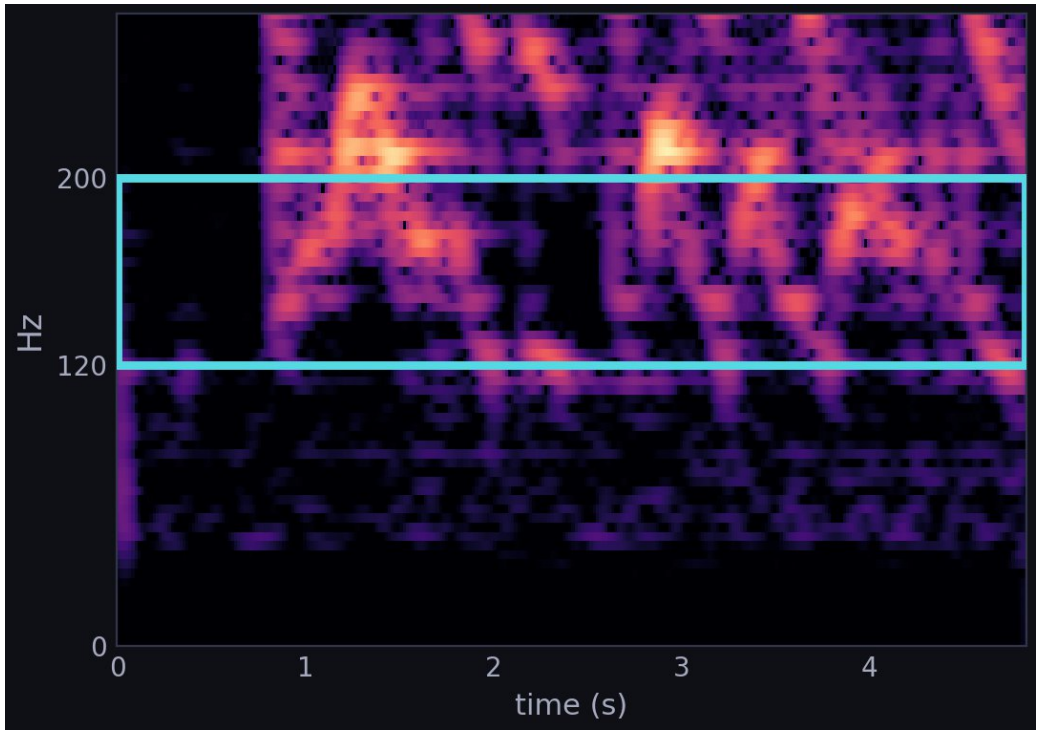
3

Adversarial Techniques and Laundering Attacks

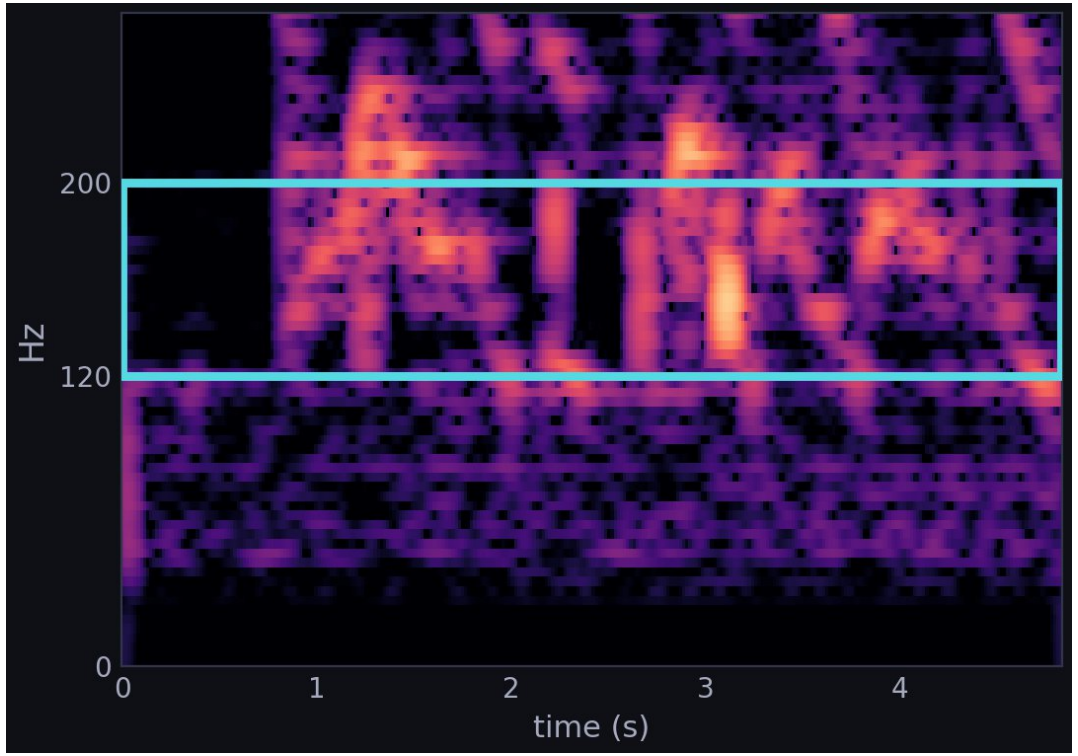
Adversarial Techniques



Voice PAD Bypass via frequency restoration



Replayed

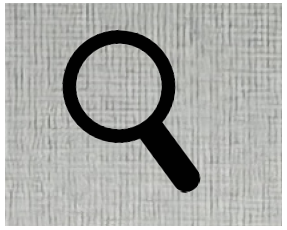


Restored

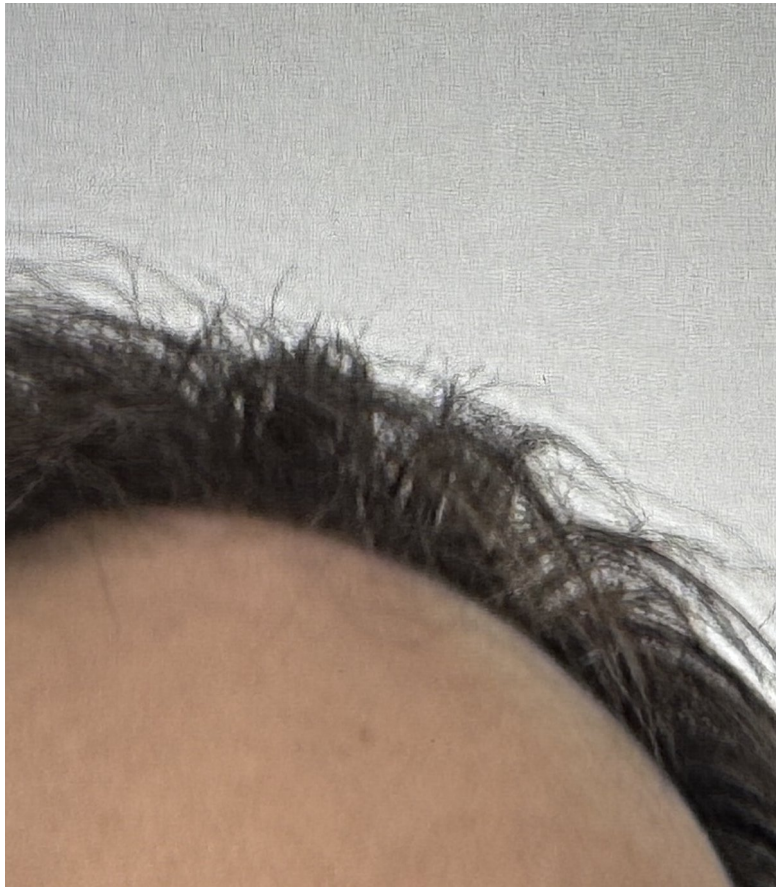
Wang et al., *When the Differences in Frequency Domain are Compensated: Understanding and Defeating Modulated Replay Attacks on Automatic Speech Recognition*, ACM CCS 2020.

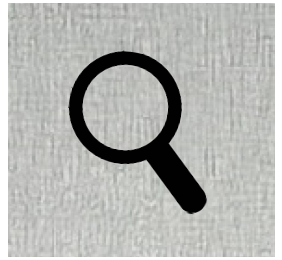
Face PAD Evasion – Demoire Attack in Action





Before





After

Recaptured Raw Screen Image and Video Demoiréing via Channel and Spatial Modulations https://arxiv.org/pdf/2310.20332

VIDEO

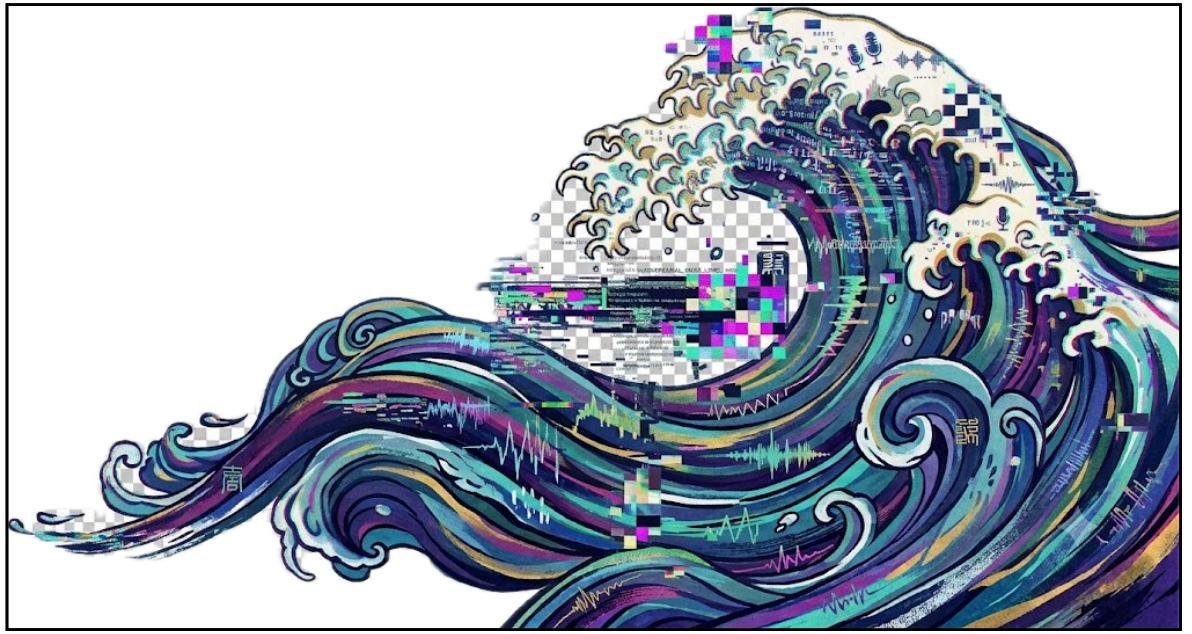
Adversarial Techniques

Visual Perturbations

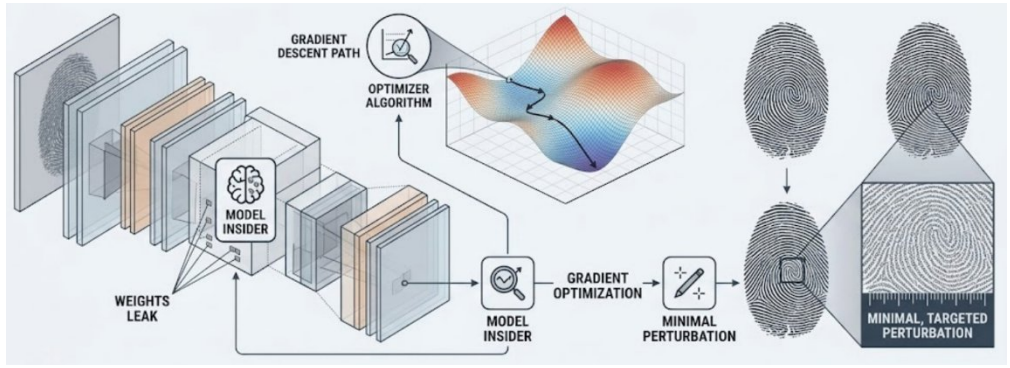


https://www.researchgate.net/figure/A-single-Universal-Adversarial-Perturbation-35-can-fool-a-model-on-multiple-images_fig5_353654246

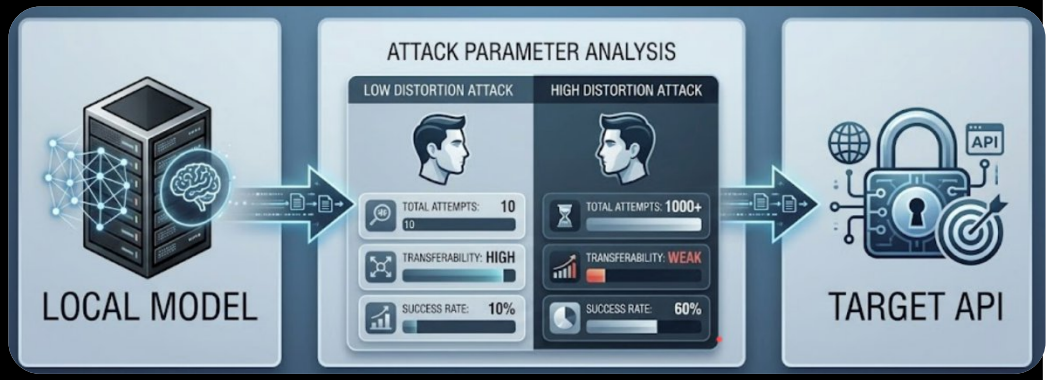
Audio spectrum perturbations



Black-Box vs WhiteBox Adversarial Techniques

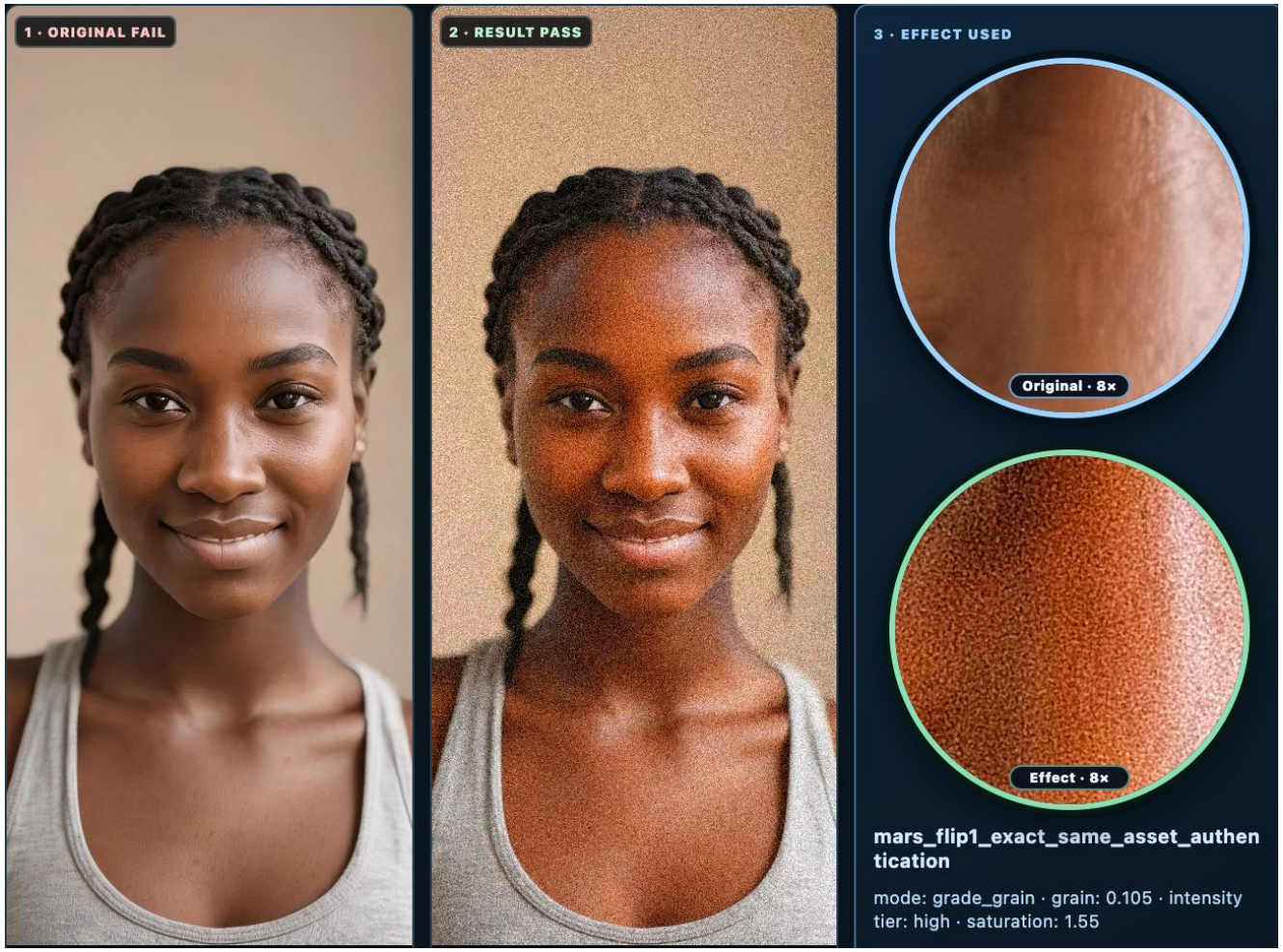


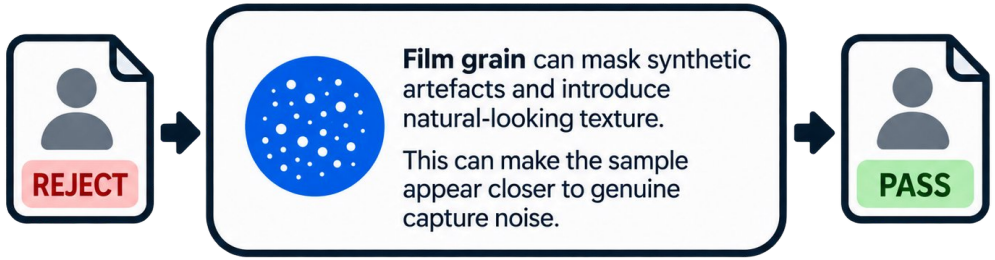
- *Gradient-driven optimisation*
- *Minimal, precisely targeted perturbations*
- *Model-insider, weights leak or SDK disclosures*



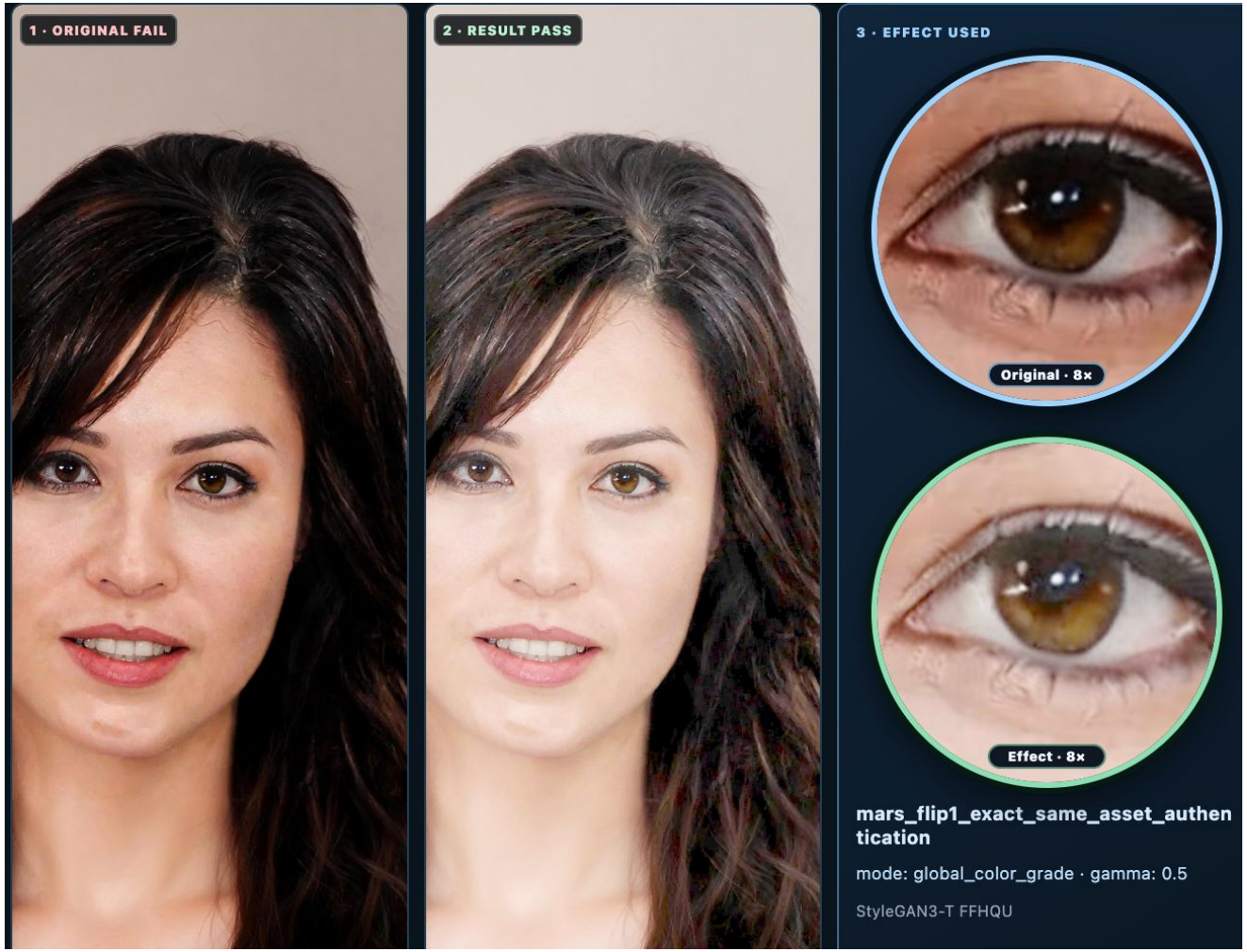
- *Query-based search or transfer from a surrogate model*
- *Higher distortions, many attempts*
- *Models a realistic attempt against an API*

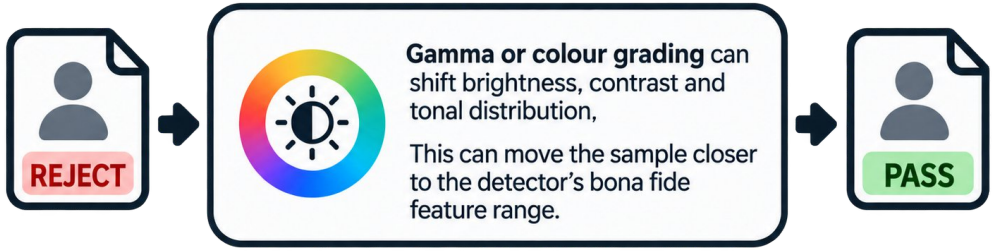
Synthetic Detection – Grain Effect





Synthetic Detection – Brightness tweak





Synthetic Detection – MARS Adversarial Technique

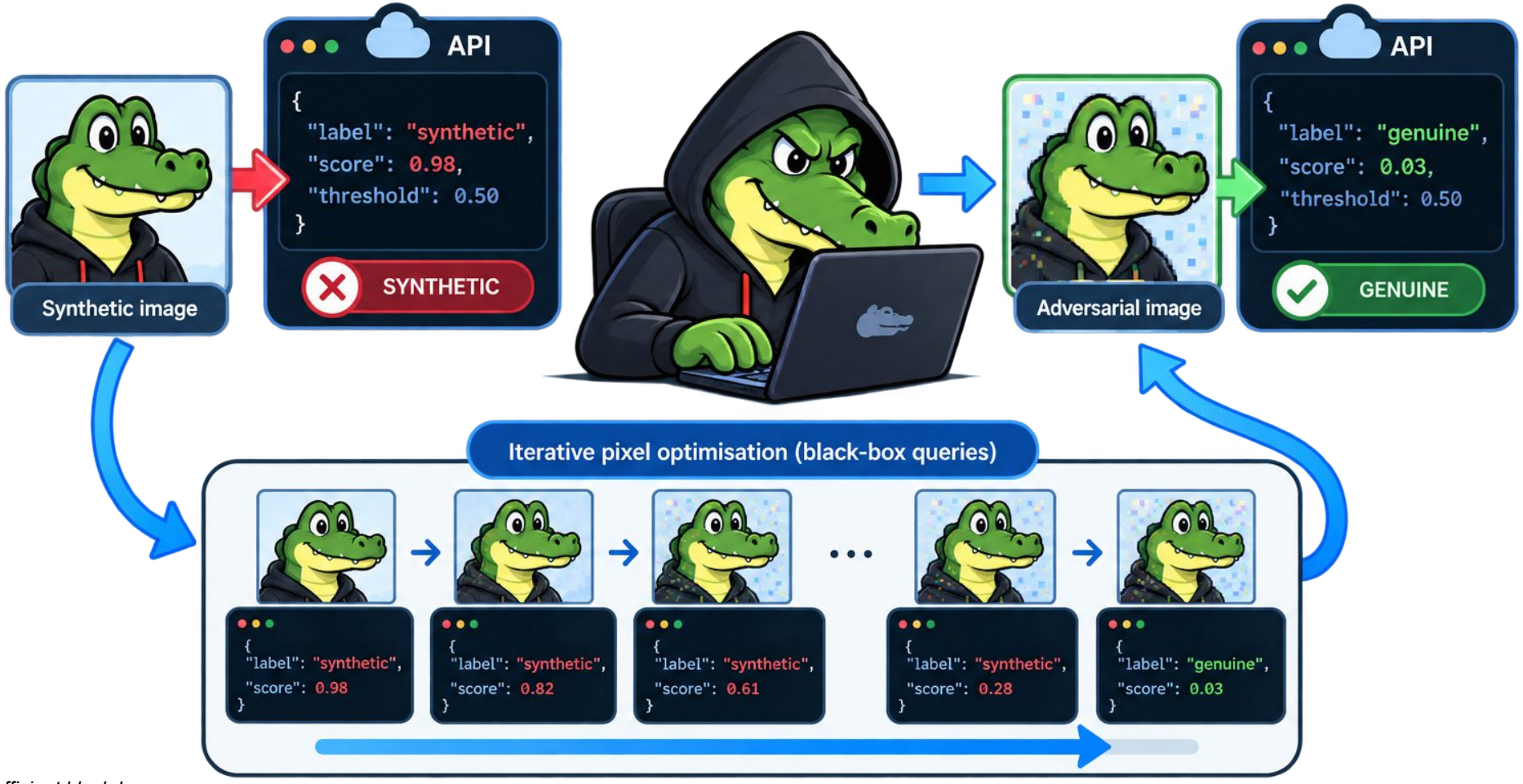


MARS
Multi-detector Authentic Reference-set Steering



MARS makes tiny, bounded changes to a synthetic image to steer its statistical representation toward genuine reference images across several surrogate detectors, while keeping the face and image visually almost unchanged.

Black-Box Square Attack



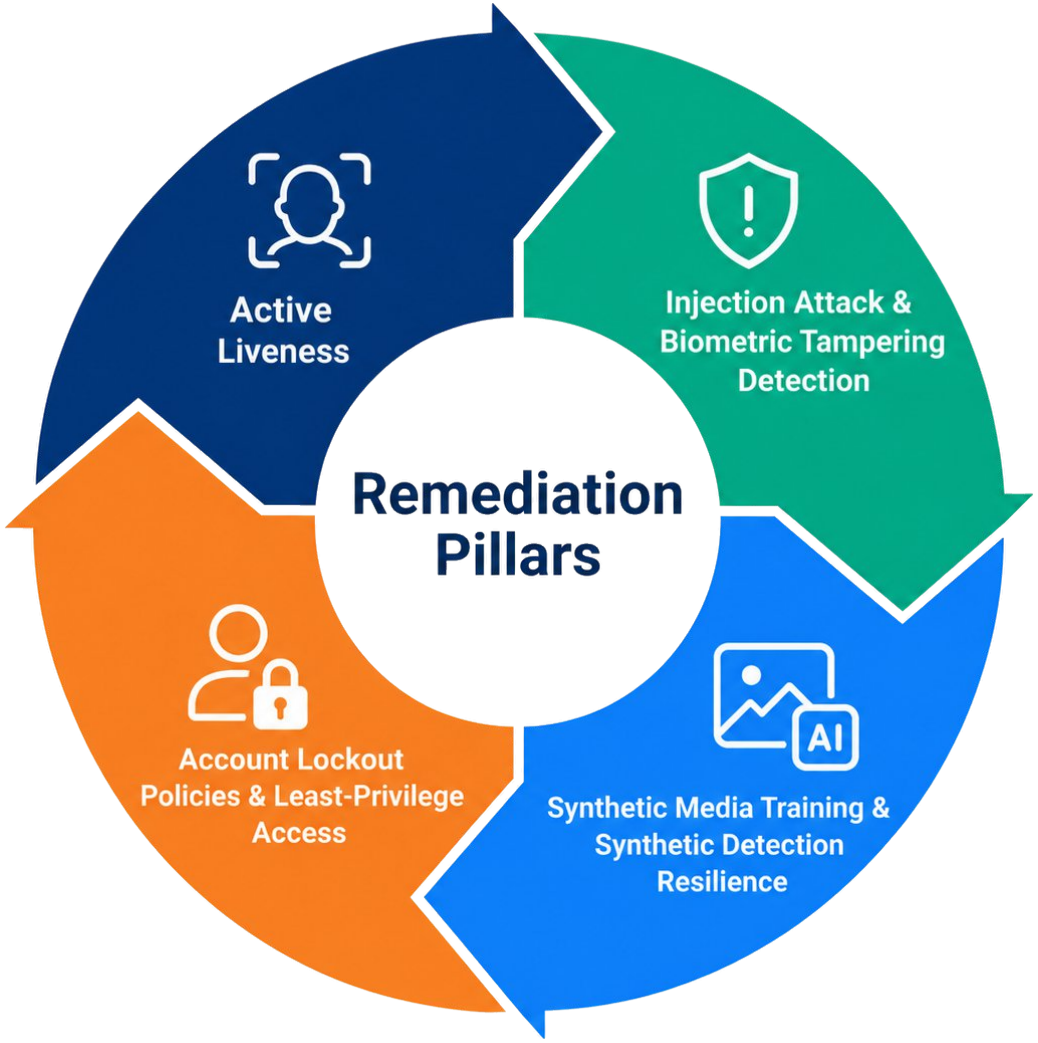
Square Attack: a query-efficient black-box adversarial attack via random search https://arxiv.org/pdf/1912.00049

DEMO

4

Main Defences

Remediation Plan



Additional Remediations



IVR Gating in Mobile Apps

Restrict access to IVR-sensitive workflows in mobile applications when Injection Attack Detection (IAD) is enabled.



R&D on Image Authenticity Attestation

Evaluate emerging features that attest the authenticity and provenance of captured images.

E.g. Newly released, for the US market, **Apple Reference Image** which may prove helpful in thwarting injection attacks on iOS devices. Ineffective for voice though.

https://security.apple.com/blog/apple-reference-image/

Questions?





AlligatorCon Europe 2026

secforce.com